

SkillClaw: Collective Skill Evolution

Let LLM agent skills evolve continuously – aggregating cross-user interaction trajectories and applying agentic, evidence-driven skill updates across multi-user agent ecosystems.

Multi-User LLM Agents

Agentic Evolver Pipeline

Collective Skill Evolution

WildClawBench · Qwen3-Max

Agent Skills Remain Static After Deployment

✗ STATIC PARADIGM — TODAY

Skills Frozen at Install

- Skills manually installed from a centralized hub
- Solutions discovered per-session never persist
- Same failure patterns re-encountered by every user
- Each user rediscovers solutions independently
- No system-level knowledge accumulation occurs

✓ EVOLVING PARADIGM — SKILLCLAW

Skills Improve from Runtime Evidence

- Interaction trajectories aggregated across users
- Cross-user patterns expose skill behavioral limits
- Agentic evolver diagnoses and proposes mutations
- Verified skill updates merged into shared repository
- All agents synchronized — knowledge compounds

Closed-Loop Pipeline: Interaction → Collection → Evolution → Distribution



Three-Stage Pipeline: Evidence → Attribution → Evolution

01

Evidence

Analyze structured session trajectories grouped by the referenced skill. Identify behavioral patterns: recurring failure modes, success sequences, and edge conditions exposed by diverse user contexts.

→ Trajectory pattern analysis

02

Attribution

Diagnose root causes of observed failures and successes. Determine whether issues stem from specification gaps, missing edge-case handling, or incorrect assumptions in the current skill implementation.

→ Root cause diagnosis

Key insight: Evolution is driven by an autonomous agent performing open-ended reasoning over interaction evidence – not predefined update rules. This enables context-aware mutations that adapt to novel failure modes.

Cross-User Aggregation Exposes Skill Behavioral Boundaries

Compatible Claw-style agent systems contributing trajectories:

OpenClaw CoPaw IronClaw PicoClaw ZeroClaw NanoClaw NemoClaw

// Different users exercising the same skill under diverse contexts produce **complementary behavioral views** - revealing both conditions under which it works and those under which it breaks

// Trajectories preserve full **action-feedback causal chains**, enabling precise attribution of failures to specific skill steps rather than user behavior

// Aggregated evidence covers a **wider behavioral distribution** than any single user's sessions - multiple users reveal the true boundary of a skill's applicability

Multimodal Creative Pipeline: 6-Day Evolution Log

Candidates: 6 Accepted: 1 Rejected: 5 Acceptance rate: 16.7%

DAY	CANDIDATE SKILL	SKILL FUNCTION	VALIDATOR	NEXT-DAY ACTION
Day 1	validate-tmp-workspace-inputs	Check /tmp_workspace inputs & environment setup	ACCEPT	Upgrade to new best pool
Day 2	multimodal-input-validation-and-setup	Multimodal input validation & output env init	REJECT	Keep current best pool
Day 3	multimodal-validation-and-setup	Multimodal input validation & output env init	REJECT	Keep current best pool

1 of 6 candidates accepted — conservative verification prevents regressions. Only Day 1's workspace validation improvement cleared the bar; all subsequent creative-pipeline mutations were rejected.

06 - CASE STUDY B

Git Push Auth-Fallback Skill: Iterative Refinement Over 6 Days

Candidates: 6 Accepted updates: 3 Rejected: 2 Acceptance rate: 50%



Three Properties That Distinguish SkillClaw

01

Collective Evolution

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem. No user's runtime discoveries are discarded after their session ends.

02

Fully Automatic

Skill evolution is driven entirely by runtime interaction – no manual curation, no explicit user intervention. The loop from evidence to distribution closes autonomously.

03

Agentic Evolution Paradigm

Skill updates are produced through open-ended agent reasoning over evidence, not predefined update rules. The evolver adapts to novel failure modes and emergent behavioral patterns.